

Question:

If the parabola $y_1 = x^2 + 2x + 7$ and the line $y_2 = 6x + b$ intersect at only one point, what is the value of b ?

Answer Choices:

- A. 10
- B. 5
- C. 3
- D. 12
- E. 4
- F. 6
- G. 8
- H. 7
- I. 2
- J. 9

Reasoning + Answer:

<think> For the parabola and line to intersect at only one point, the line must be tangent to the parabola. This means that the discriminant of the quadratic equation formed by equating the two must be zero. The equation of the parabola is $y = x^2 + 2x + 7$, and the equation of the line is $y = 6x + b$. Setting the two equations equal to each other, we get $x^2 + 2x + 7 = 6x + b$. Rearranging the terms, we get $x^2 - 4x + 7 - b = 0$. The discriminant of this quadratic equation is $(-4)^2 - 4(1)(7 - b)$. Setting the discriminant equal to zero, we get $16 - 4(7 - b) = 0$. Solving for b , we get $b = 3$. </think>
><answer> 3 </answer>

